

RDST.jl — Streamable Random Number Generators

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RDST.jl Documentation

Streamable pseudo-random number generators for Julia: MRG32k3a (L'Ecuyer) and xoshiro256+ (Blackman & Vigna), with non-overlapping streams and substreams.

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Quick example

using RDST

```

gen  = MRG32k3aGen()
rngA = next_stream(gen)      # independent stream A
rngB = next_stream(gen)      # independent stream B

rand(rngA)                   # Float64 in [0, 1)
next_substream!(rngA)        # move to the next substream of A
reset_stream!(rngA)           # rewind stream A to its beginning

```

Getting Started

RDST.jl provides pseudo-random number generators with support for **non-overlapping streams and substreams**, following the stream/substream model of L'Ecuyer et al. (2002).

Installation

using Pkg

Pkg.add(url = "https://github.com/wirelessroom2/RDST.jl")

or, from a local clone:

using Pkg

Pkg.develop(path = "path/to/RDST.jl")

Requirements: Julia ≥ 1.6 . The only dependency is the standard library Random.

Choosing a generator

RDST.jl implements the full xoshiro/xoroshiro family of Blackman & Vigna plus L'Ecuyer's MRG32k3a. All variants of a xoshiro family share the same linear transition and jump constants; only the output scrambler differs.

Type	Family	State	Period	Scrambler
Xoroshiro128p	xoroshiro128	128 bits	$2^{128} - 1$	$s_0 + s_1$
Xoroshiro128ss	xoroshiro128	128 bits	$2^{128} - 1$	high bits, **
Xoroshiro128pp	xoroshiro128	128 bits	$2^{128} - 1$	high bits, ++
Xoshiro256p	xoshiro256	256 bits	$2^{256} - 1$	$s_0 + s_3$
Xoshiro256ss	xoshiro256	256 bits	$2^{256} - 1$	high bits, **
Xoshiro256pp	xoshiro256	256 bits	$2^{256} - 1$	high bits, ++
Xoshiro512p	xoshiro512	512 bits	$2^{512} - 1$	$s_0 + s_2$
Xoshiro512ss	xoshiro512	512 bits	$2^{512} - 1$	high bits, **
Xoshiro512pp	xoshiro512	512 bits	$2^{512} - 1$	high bits, ++
MRG32k3a	combined MRG (L'Ecuyer)	192 bits	$\approx 2^{191}$	native Float64

See [Generator Comparison](#) for guidance and benchmarks.

First numbers

using RDST

```
rng = MRG32k3a()      # default seed [12345, ..., 12345]
rand(rng)              # 0.12701112204657714
rand(rng)              # 0.3185275653967945

x = Xoshiro256p(UInt64[1, 2, 3, 4])
rand(x)                # Float64 in [0, 1)
```

Supported output types

MRG32k3a

- Float64 (native), plus Float32, Float16 via conversion
- UInt8, UInt16, UInt32, UInt64, UInt128
- Int8, Int16, Int32, Int64, Int128 (same bit patterns)
- Bool
- Ranges: `rand(rng, 1:10)` works through the standard Random machinery

Note: integer outputs of MRG32k3a are built from its native ~31-bit resolution; they are not uniform over their full width. Use them for indices, choices and flags rather than cryptography.

Xoshiro256p

- Float64 (native path), Float32, Float16
- UInt64
- Ranges: `rand(rng, r::UnitRange{Int64})`

Reproducibility

Every generator accepts an explicit seed:

```
rng = MRG32k3a([42, 1, 2, 3, 4, 5])  
x    = Xoshiro256p(UInt64[0xdead, 0xbeef, 0xcafe, 0xbabe])
```

Seeds are validated by `checkseed`: a valid MRG32k3a seed has 6 non-negative values; the first three must be $< m1$ and not all zero, the last three $< m2$ and not all zero.

Next steps

- [Streams & substreams](#)
- [API reference](#)
- [Implementation notes](#)

Streams & Substreams

The central abstraction of RDST.jl is the *stream*: a long, non-overlapping subsequence of a generator's period. Streams can themselves be split into *substreams*. This is the architecture recommended by L'Ecuyer et al. (2002) for parallel and replicated stochastic simulation:

- **Streams** are handed to independent replications or parallel workers; their outputs never overlap, so replications are statistically independent.
- **Substreams** delimit scenarios *within* one replication; rewinding a substream while keeping the scenario structure enables techniques such as common random numbers.

The two object families

1. **Stream generators** (`<: AbstractRNGStream`) hold the seed of the *next* stream to be produced:
 - MRG32k3aGen
 - Xoshiro256plusGen
2. **RNGs** (`<: AbstractStreamableRNG`) actually produce numbers and can move along streams/substreams:
 - MRG32k3a
 - Xoshiro256p

Producing independent streams

using RDST

```
gen = MRG32k3aGen()           # or Xoshiro256plusGen(UInt64[...])

worker1 = next_stream(gen)     # stream 1
worker2 = next_stream(gen)     # stream 2 – guaranteed disjoint from stream 1
worker3 = next_stream(gen)     # stream 3 – ...
```

Each call to `next_stream` advances the generator's internal seed by a huge leap (2^{127} values for MRG32k3a, a full `long_jump` for Xoshiro256+), which is what guarantees non-overlap.

```
# Typical parallel loop
using Distributed
gen = MRG32k3aGen()
@distributed (+) for _ in 1:nworkers()
    rng = next_stream(gen)     # note: ship seeds explicitly in real code,
    sum(rand(rng) for _ in 1:10^6)
end
```

Navigating substreams

Within a single RNG, three functions move along the stream/substream hierarchy (all return the modified RNG):

Function	Effect
<code>reset_stream!(rng)</code>	jump to the very beginning of the current stream
<code>reset_substream!(rng)</code>	jump to the beginning of the current substream
<code>next_substream!(rng)</code>	advance to the next substream

Example — common random numbers across two scenarios:

```
rng = next_stream(MRG32k3aGen())

rand(rng)           # consume some variates of substream 1...
rand(rng)

next_substream!(rng) # switch to substream 2 for scenario B
uB = rand(rng)

reset_stream!(rng)   # replay substream 1 from scratch for scenario A
uA = rand(rng)       # same underlying uniforms as the first draw
```

For Xoshiro256+, `next_substream!` is implemented as a `short_jump` (2^{96} -value leap) and `reset_stream!/next_substream!` manipulate the saved Bg/Ig checkpoints.

Saving, restoring, jumping

```
state = get_state(rng)      # a copy; safe to store
# ... consume numbers ...
```

To restore an MRG32k3a state, rebuild the generator with the triple constructor (Cg, Bg, Ig all set to the snapshot):

```
snapshot = get_state(rng)
xs = [rand(rng) for _ in 1:5]
clone = MRG32k3a(snapshot, snapshot, snapshot)
rand(clone) == xs[1]        # true – resumes exactly after the snapshot
```

MRG32k3a additionally supports arbitrary jumps inside a stream:

```
advance_state!(rng, e, c)
```

moves the state forward by n steps, where $n = 2^e + c$ (e may be negative, and negative c moves backwards). This costs $O(\log n)$ matrix operations, independent of the distance jumped.

Guarantees

- Streams produced by successive calls to `next_stream` on the same generator object are **provably non-overlapping**.
- Substreams likewise partition a stream into disjoint blocks (length $\approx 2^{76}$ steps for MRG32k3a).

API Reference

All names below are exported by the RDST module unless marked (*internal*).

Types

AbstractRNGStream — supertype of stream *generators* (objects that mint new independent streams): MRG32k3aGen, Xoshiro256plusGen.

AbstractStreamableRNG <: **Random.AbstractRNG** — supertype of usable RNGs that can navigate streams and substreams: MRG32k3a, Xoshiro256p.

MRG32k3a

Constructors

Signature	Description
MRG32k3a()	default seed [12345, 12345, 12345, 12345, 12345, 12345]
MRG32k3a(x::Vector{Int})	seed all three states (Cg = Bg = Ig = x)

Signature	Description
<code>MRG32k3a(x, y, z::Vector{Int})</code>	explicit current/substream/stream starts

Seeds are validated with `checkseed`; invalid seeds throw an `AssertionError`.

Functions

`rand(rng::MRG32k3a) -> Float64` Uniform in $[0, 1)$ with ~ 32 bits of resolution. This is the native output.

`rand(rng::MRG32k3a, T)` Supported `T`: `Float64`, `Float32`, `Float16`, `UInt8...UInt128`, `Int8...Int128`, `Bool`, and any `UnitRange` (e.g. `rand(rng, 1:10)`).

`reset_stream!(rng) -> rng` Set the current state (`Cg`) *and* the substream start (`Bg`) back to the stream start (`Ig`).

`reset_substream!(rng) -> rng` Set `Cg` back to `Bg`.

`next_substream!(rng) -> rng` Advance `Bg` to the next substream boundary (jump of $\approx 2^{76}$ steps) and set `Cg = Bg`.

`advance_state!(rng, e::Integer, c::Integer)` Jump forward/backward inside the current substream by `n` steps where $n = 2^e + c$. Negative `e` or `c` move backwards. Cost is $O(\max(|e|, \log |c|))$ matrix operations.

`get_state(rng) -> Vector{Int}` Return a copy of `Cg`.

`checkseed(x) -> Bool` true iff `x` has length 6, all entries ≥ 0 , entries 1-3 are $< m1$ and not all zero, entries 4-6 are $< m2$ and not all zero.

`DEFAULT_SEED` The constant seed vector used by `MRG32k3a()` and `MRG32k3aGen()`.

`copy(m::MRG32k3a) -> MRG32k3a` Independent deep copy of the generator (all three state vectors).

Stream generation

Signature	Description
<code>MRG32k3aGen()</code> , <code>MRG32k3aGen(seed)</code>	create a stream generator
<code>next_stream(gen) -> MRG32k3a</code>	return a new stream; internal seed leaps 2^{127} values ahead
<code>get_state(gen) -> Vector{Int}</code>	seed that will be used by the next <code>next_stream</code> call

Xoshiro256+

Constructors

Signature	Description
<code>Xoshiro256p(s::NTuple{4,UInt64}) /</code> <code>Xoshiro256p(v::Vector{<:Unsigned})</code> <code>Xoshiro256p(x, y, z)</code>	<code>Cg = Bg = Ig = s</code> explicit current/substream/stream states

Functions

rand(rng::Xoshiro256p) -> Float64 Uniform in [0, 1), built from a full 64-bit draw.

rand(rng::Xoshiro256p, T) Supported T: Float64, Float32, Float16, UInt64, and `UnitRange{Int64}` (uniform, unbiased modulo sampling).

srand(rng, seed::Vector{UInt64}) -> rng Re-seed an existing generator (first 4 words are used).

reset_stream!(rng) -> rng, reset_substream!(rng) -> rng, next_substream!(rng) -> rng Same semantics as for MRG32k3a; `next_substream!` performs a `short_jump` (leap of $\approx 2^{96}$ values).

short_jump(rng) -> rng Jump to the beginning of the next 2^{96} -value block (Vigna's short jump).

long_jump(rng) -> rng Leap $\approx 2^{192}$ values ahead — used to delimit streams.

get_state(rng) -> Vector{UInt64} Copy of the current state `Cg`.

copy(rng) -> Xoshiro256p Independent copy.

(internal) **next(rng) -> UInt64**: raw 64-bit output; also available as `rand(rng, UInt64)`.

Stream generation

Signature	Description
<code>Xoshiro256plusGen(seed::Vector{UInt64})</code>	create a stream generator (seed must have exactly 4 words)
<code>next_stream(gen) -> Xoshiro256p</code>	new independent stream (<code>long_jump</code> from the stored seed)
<code>srand(gen, seed::Vector{UInt64}) -> gen</code>	reset the stored seed
<code>get_state(gen) -> Vector{UInt64}</code>	seed that will be used by the next <code>next_stream</code> call

Xoshiro / xoroshiro families

All variants share one implementation (LinRNG{N,S} internally, N = state words, S = scrambler) and expose the same API as **Xoshiro256+** above. Exported types:

RNGs	Stream generators
Xoroshiro128p, Xoroshiro128ss, Xoroshiro128pp	Xoroshiro128pGen, Xoroshiro128ssGen, Xoroshiro128ppGen
Xoshiro256p, Xoshiro256ss, Xoshiro256pp	Xoshiro256plusGen, Xoshiro256ssGen, Xoshiro256ppGen
Xoshiro512p, Xoshiro512ss, Xoshiro512pp	Xoshiro512pGen, Xoshiro512ssGen, Xoshiro512ppGen

Constructors accept an NTuple{N,UInt64} or a Vector{<:Unsigned} of length N (1 or 3 arguments). Available functions: rand (Float64/Float32/Float16, UInt64, ranges), srand, reset_stream!, reset_substream!, next_substream!, short_jump, long_jump, get_state, copy.

advance_state!(rng, e::Integer, c::Integer) -> rng Jump forward by n steps ($n = 2^e + c$ if $e > 0$, $n = -2^{(-e)} + c$ if $e < 0$, $n = c$ if $e = 0$) or backward if $n < 0$. Implemented by computing the jump polynomial $x^n \bmod p(x)$ over GF(2) (p = the family's characteristic polynomial) and applying it to the current state; distances are reduced modulo the period. Only Cg moves — stream/substream boundaries are kept. Cost is $O((64N)^2)$ GF(2) operations (≈ 10 –500 ms depending on family); for the standard distances prefer short_jump/long_jump, which use precomputed constants and are allocation-free.

Jump semantics (all xoshiro-family generators):

- short_jump(rng) — jump anchored at the current **substream start** (Bg); lands at the next substream boundary (2^{64} values for xoroshiro128, 2^{128} for xoshiro256, 2^{256} for xoshiro512).
- long_jump(rng) — jump anchored at the **stream start** (Ig); delimits independent streams.
- next_stream(gen) applies the long-jump polynomial to the stored seed.

All jump constants are the official ones from xoshiro.di.unimi.it and outputs are validated byte-for-byte against the original C implementations.

Method index

Function	MRG32k3a	Xoshiro256p
rand (Float64)	yes	yes
rand (other floats/ints/bool)	yes	partial
rand (range)	yes	yes (UnitRange{Int64})
reset_stream!	yes	yes

Function	MRG32k3a	Xoshiro256p
reset_substream!	yes	yes
next_substream!	yes	yes
advance_state!	yes	—
srand	—	yes
get_state	yes	yes
next_stream (via Gen)	yes	yes

Implementation Notes

MRG32k3a

MRG32k3a is L'Ecuyer's combined multiple recursive generator of order 3. It maintains two 3-component integer states updated by the recurrences

$$\begin{aligned}
 p1 &= (1403580 \cdot x2 - 810728 \cdot x1) \bmod m1, & m1 &= 2^{32} - 209 \\
 p2 &= (527612 \cdot y2 - 1370589 \cdot y0) \bmod m2, & m2 &= 2^{32} - 22853 \\
 u &= (p1 - p2) / m1 \quad (\text{with wrap-around}) \in [0, 1)
 \end{aligned}$$

The combined generator has period $\approx 2^{191}$.

Modular arithmetic without overflow

All state components stay below $m1 < 2^{32}$, so products fit in Float64 exactly ($< 2^{53}$). The helper `MultModM(a, s, c, m)` computes $(a \cdot s + c) \bmod m$ using this fact and falls back to a split-by- 2^{17} computation when intermediate values would exceed 2^{53} . Matrix helpers:

- `MatVecModM(A, s, m)` — matrix-vector product mod m
- `MatMatModM(A, B, m)` — matrix product mod m
- `MatTwoPowModM(A, e, m)` — A raised to the power 2^e
- `MatPowModM(A, n, m)` — A^n by binary exponentiation

Streams, substreams, jumps

Moving a stream forward by k steps is a linear operation on its state, hence a precomputed matrix power applied to the state vector:

Operation	Matrix	Step
<code>next_stream(gen)</code>	<code>A1p127, A2p127</code>	2^{127} values
<code>next_substream!(rng)</code>	<code>A1p76, A2p76</code>	2^{76} values
<code>advance_state!(rng, e, c)</code>	<code>MatTwoPowModM / InvA1, InvA2</code>	arbitrary (inverse matrices allow backward jumps)

The RNG keeps three checkpoints: C_g (current), B_g (substream start), I_g (stream start), which is what makes `reset_substream!` and `reset_stream!` $O(1)$.

Xoshiro256+

xoshiro256+ (Blackman & Vigna, 2019) keeps four 64-bit words and produces one word per step with only shifts, XORs and rotations:

```
result = s1 + s4
t = s2 << 17
s3 = xor(s3, s1); s4 = xor(s4, s2); s2 = xor(s2, s3); s1 = xor(s1, s4); s3 = xor(s3
```

The output word is the sum of two internal words (+ variant): fastest of the xoshiro family for float generation, though the low three bits of the raw output have limited linear complexity — irrelevant once scaled to Float64 (53 bits used). Period: $2^{256} - 1$.

Jumps

short_jump and long_jump use Vigna’s published jump polynomials. They are computed as XOR-linear combinations of states visited while stepping through the 256 bits of each jump constant. RDST.jl hoists the jump constants into compile-time tuples and evaluates jumps allocation-free on immutable state tuples.

RDST semantics differ subtly from the reference C code: jumps are **anchored** at stream boundaries. short_jump(rng) first resets $C_g \leftarrow B_g$, then applies the jump polynomial, and stores the result in both C_g and B_g ; long_jump(rng) does the same with respect to I_g . This matches L’Ecuyer’s stream model (next_substream!/reset_stream!) and makes scenario replay reproducible regardless of consumption inside the current substream.

Families and scramblers

The package implements all 64-bit variants from xoshiro.di.unimi.it through a single generic type LinRNG{N,S} (state words $\times$ scrambler):

- transition _lin_step for NTuple{2} (xoroshiro128: a=24, b=16, c=37), NTuple{4} (xoshiro256) and NTuple{8} (xoshiro512);
- scramblers +, ** (rotr(s2·5,7)·9) and ++ (rotr(s1+s4,23)+s1 for 256; family-specific constants);
- per-family jump constants shared by all scramblers.

Exported aliases: Xoroshiro128p/ss/pp, Xoshiro256p/ss/pp, Xoshiro512p/ss/pp plus matching *Gen stream generators.

All nine variants are regression-tested against byte-exact outputs produced by the original C implementations compiled with gcc (sequences, short jumps and long jumps).

Arbitrary forward/backward jumps

Because the transition is multiplication by x in $\text{GF}(2)[x]/p(x)$ — with p the family’s characteristic polynomial, as published in Vigna’s reference files — one can jump any distance n in constant time by applying the polynomial $x^n \bmod p$ through the

same accumulate-and-step loop as the fixed jumps. Backward distances use the multiplicative order of x : $x^{(-n)} = x^{(2^{\text{deg}}-1-n)}$ ($\text{deg} = 64N$). RDST.jl implements a small GF(2) polynomial engine (`_poly_mul_mod`, `_poly_pow_x`, BigInt exponents reduced modulo the period) and exposes it as `advance_state!(rng, e, c)` with the exact distance convention of MRG32k3a. Correctness is property-tested: fixed jumps coincide with `short_jump`, round trips (+k then -k) restore the state bit-for-bit, and backward-jumped generators re-emit previously seen values.

State representation

The state lives in `NTuple{4, UInt64}` fields. Immutable tuples let the JIT keep the whole working set in registers inside `next`, eliminating bounds checks and heap traffic that a `Vector{UInt64}` representation incurs. Measured throughput: $\approx 10^9$ draws/s on a single core.

Integer outputs of MRG32k3a

MRG32k3a natively yields ~ 31 bits per draw (the difference $p_1 - p_2$). Wider unsigned types are assembled from 16-bit chunks of that value; narrower types are truncations. Consequence: these integers are *not* uniform over their full width — fine for indexing/shuffling/flags in simulations, not for cryptographic use.

Testing strategy

Reference sequences were captured from the reference implementations (LEcuyer’s C code semantics for MRG32k3a; Vigna’s constants for the xoshiro jumps) and regression-checked after optimization: identical outputs are produced for every supported type, plus reset/jump/state round-trips.

Performance snapshot

Single core, Julia 1.12 (indicative):

Operation	Throughput
<code>rand(::MRG32k3a)</code>	$\approx 200 \times 10^6/\text{s}$
<code>rand(::Xoroshiro128p)</code>	$\approx 1200 \times 10^6/\text{s}$
<code>rand(::Xoshiro256p/ss/pp)</code>	$\approx 1340 \times 10^6/\text{s}$
<code>rand(::Xoshiro512p/ss/pp)</code>	$\approx 1150 \times 10^6/\text{s}$
<code>short_jump / long_jump</code>	0 allocations

Comparing generators: xoshiro variants vs MRG32k3a

RDST.jl lets you pick between two very different generator designs. This page helps you choose.

Side-by-side

Property	MRG32k3a	Xoroshiro128* (p/ss/pp)	Xoshiro256* (p/ss/pp)	Xoshiro512* (p/ss/pp)
State (bits)	192 (6 × Int64)	128	256	512
Period	$\approx 2^{191}$	$2^{128} - 1$	$2^{256} - 1$	$2^{512} - 1$
Native output	Float64 [0,1), ~32-bit resolution	UInt64	UInt64	UInt64
Throughput (this package, single core)	≈ 200 M/s	≈ 1200 M/s	≈ 1340 M/s	≈ 1150 M/s
Stream mechanism	matrix power $A^{2^{127}}$ on the seed	long_jump polynomial (2^{96})	long_jump (2^{192})	long_jump (2^{384})
Substream mechanism	matrix power $A^{2^{76}}$	short_jump (2^{64})	short_jump (2^{128})	short_jump (2^{256})
Backward jumps	yes (advance_state!)	yes (advance_state!, GF(2) polynomials)	yes (advance_state!)	yes (advance_state!)
Arbitrary-jump cost	O(e) matrix products	O(deg ²) GF(2) ops (~10-500 ms)	same	same
Equidistribution	well analysed theory	1-dim (**/++) / 0-dim (+)	3-dim (**/++) / 2-dim (+)	7-dim (**/++) / 6-dim (+)
BigCrush (Vigna's shootout)	passes	passes	passes	passes
Parallelism scale advised	large	mild ($\approx 2^{32}$ streams)	very large	extreme

(* means any of the +, **, ++ scramblers.)

When to choose MRG32k3a

- You need **provably non-overlapping streams with published, peer-reviewed parameters** (L'Ecuyer et al. 2002), e.g. to stay compatible with simulation libraries built on that model (SSJ, Simio, Arena-style stream ids...).
- Your application consumes **floats only**: MRG32k3a emits them natively.
- You can afford ~6x slower generation.

Note: backward jumping is *no longer* a differentiator — every RDST generator supports `advance_state!(rng, e, c)` with negative distances.

When to choose a xoshiro/xoroshiro variant

- You want **maximum throughput** (all are $\approx 6\text{--}7\times$ faster than MRG32k3a here) or raw 64-bit integers.
- Choose your state size by parallelism needs:
 - Xoroshiro128*: embedded/GPU or few workers only — its 2^{64} substreams per 2^{32} streams are enough for mild parallelism. Note Xoroshiro128p has a mild Hamming-weight dependency after ~ 5 TB of output; prefer ss/pp.
 - Xoshiro256*: the sweet spot for general use (Julia itself uses xoshiro256++ as its default RNG). Use + if you generate *only* floating-point numbers from the high bits ($\sim 15\%$ faster historically); otherwise prefer ++/**, whose outputs are fully equidistributed.
 - Xoshiro512*: essentially never needed for period reasons — any period $\geq 2^{256}$ is beyond every imaginable need — but handy when you want many more independent substreams than 2^{128} without changing design.

Scrambler choice within a family

Scrambler	Output	Best for
+	sum of two state words	float-only workloads using the top bits; lowest 3 bits have low linear complexity
**	<code>rotr(s2·5, 7)·9</code>	all-purpose; maximal equidistribution (e.g. .NET/Lua default)
++	<code>rotr(s1+s4, k)+s1</code>	all-purpose; Rust's <code>SmallRng</code> , Julia's default

All three share the same transition and therefore the same jump constants: streams/substreams behave identically across scramblers of one family.

Practical notes

- Seeding: seed states must not be all-zero. For hash-like seeds, initialize with `splitmix64` outputs (see Vigna's recommendation).
- The package's jumps are **anchored at stream boundaries** ($C_g \leftarrow B_g \leftarrow I_g$), matching the L'Ecuyer stream model: `next_substream!` always lands at the same place regardless of how far you consumed in the current substream, which makes scenario replay reproducible.